

Jeffrey William Shorthill

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SUMMARY

Independent AI interpretability and evaluation researcher. Seven Zenodo-archived preprints on mixture-of-experts routing, evaluation contamination, and safety behavior, using preregistered, frozen-rule methods; published self-corrections included. Progressive Insurance's first AI Prompt Engineer. Shipped a dual-store consumer AI app and six deployed AI tools as sole engineer.

SELECTED RESEARCH (INDEPENDENT, PUBLIC, REPRODUCIBLE)

- **In-Context Availability Reorganizes MoE Expert Routing** — cross-generation study (Qwen3.5 + 3.6-35B). Literal in-context availability shifts expert routing; the attention-retrieval mechanism is located and severed; the result replicates across a model generation. Implication: multi-turn evaluation contamination. doi:10.5281/zenodo.21892999
- **When Routing Entropy Tracks Length, Not Complexity** — self-refutation of a position confound in routing-entropy analyses; the “complexity gradient” reverses under correct measurement (DeepSeek V3.1, Qwen3.5-397B). doi:10.5281/zenodo.20779602
- **The Generation Half** — expert specialization is invisible in prefill and emerges during generation; rebuts a widely cited negative result; replicates at 122B. doi:10.5281/zenodo.20779604
- **No Single Safety Gate** — harm-refusal routing is distributed; suppressing the lead expert reallocates with zero harmful completions. doi:10.5281/zenodo.20786891
- **Dialect-Marked Response Audit** — matched-pair AAVE/AE safety-support equivalence; risk signal carried by register and action lexis. doi:10.5281/zenodo.20449546
- **Register-readout artifact bundle** (e114-artifacts v1.3.1) — preregistered sweeps that retired the program's earlier claims; raw router archives, frozen specs, and SHA-256 manifests public. doi:10.5281/zenodo.20709736
- **Methods** — preregistration, hash-frozen analysis rules, blinded and sham-controlled protocols, cross-runtime validation. Includes a fully blinded gravitational-wave echo search with positive-control gates.

EXPERIENCE

Elevated Consultants LLC — Owner / Principal

Feb 2024 – present

AI product consultancy; primary client: Gift Connect (nonprofit parenting program)

Salt Lake City, UT

- Shipped the Gift Connect Parenting Suite (iOS + Android, live on both stores) and six deployed AI tools: realtime voice agent (OpenAI Realtime API/WebRTC), song and story generators, Twilio SMS assistant with CDC milestone tools, GA4 analytics dashboard, cohort portal.
- Sole engineer across mobile (React Native/Expo), web (Next.js), backends (Express, Flask/Celery), and infra (Vercel, Heroku, Firebase, Postgres), with consent-gated analytics.

Progressive Insurance — AI Prompt Engineer (2023; the company's first)

2018 – Dec 2023

Prior role: Customer Service Downtime Specialist (2018–2022)

- Proposed AI initiatives directly to executive leadership after GPT-4's release; a dedicated prompt-engineering role was created in response.
- Established internal prompting standards and taught applied AI sessions for data scientists, architects, and developers.

Webster Public Relations — Studio Manager, A/V Technical Specialist

2014 – 2018

Press-focused audio/video production

Nashville, TN

EDUCATION & SKILLS

Full Sail University — A.S., Recording Engineering (2014) • York County CC — General Studies, Dean's List

Research: experiment design, preregistration, bootstrap inference, routing/activation capture (custom llama.cpp + HF hooks), Jacobian-lens readouts, SAE probes, intervention studies • **Engineering:** Python, PyTorch, TypeScript, React/React Native, Swift/SwiftUI, Next.js, CUDA-cloud ops (Vast.ai), OpenAI/Anthropic APIs, agent frameworks